

Bibek Bhatta

bb3017@msstate.edu / +1 571-602-3140



Summary

Mechanical Engineering undergraduate with 6 plus years of experience applying **Python**, **machine learning**, and **computer vision** to **engineering** and **robotics** focused problems. Strong background in **electronics**, **3D printing**, and **embedded systems**, with hands on experience in **perception**, **data driven modeling**, and **control-oriented systems**. Experienced with **FastAPI**, **PyTorch**, **OpenCV** and **scientific computing**, and highly interested in **robotics**, **UAV systems**, **aerodynamics**, and **autonomous control system design**.

Education

Bachelor of Science in Mechanical Engineering

December 2028

Mississippi State University • Starkville, Mississippi • Sophomore • Cumulative GPA: **4.0/4.0**

- **Colvard Future Leader** Scholarship | **Freshman Academic Excellence** Scholarship | **Presidents List** Summer & Fall 2025

Cambridge International Advanced Levels

August 2023

St. Xavier's College • Maitighar, Kathmandu

- **Gold** Provincial Information Technology Olympiad | **Silver** National Information Technology Olympiad
- **Bronze** Honor International Youth Math Challenge | **Bronze** Budanilkantha National Math League
- **Bronze** National Science Exhibition

Experience

Undergraduate Researcher

Department of Plant and Soil Sciences at Mississippi State University / Starkville, MS

October 2025 - Present

- Develop a real time **ArUco** calibrated **computer vision** seed measurement system and modular Python application, supporting CNN based seed and nodule analysis for PhD research
- First author on a paper accepted for oral and poster presentation at the **90th Mississippi Academy of Sciences Conference 2026**

Technical Lead

Mathematical Association of Nepal / Kathmandu, Nepal

September 2022 - Present

- Develop comprehensive registration and training software using **Python**, **FastAPI**, **DaisyUI**, and **SvelteKit**, significantly enhancing user experience and operational efficiency
- Create a custom **PDF-to-image generator** for **LaTeX processing** and design a specialized **MongoDB database** for mathematical problems

Backend Engineer and Machine Learning Developer

usjany / Kathmandu, Nepal

August 2024 - June 2025

- Developed a production-level backend server using **FastAPI**, with **Python** libraries including **BeautifulSoup** and **Requests**, and a **PostgreSQL database**
- Built and fine-tuned a custom **Large Language Model** powered by Meta AI's **Llama-3.1 8B**, implementing a **Retrieval-Augmented Generation (RAG)** pipeline to enhance response accuracy

Machine Learning Developer

Agrisense / Kathmandu, Nepal

February 2023 - June 2023

- Developed and implemented a custom **CNN-based image processing model** utilizing **Scikit-learn**, **NumPy**, and **OpenCV** for plant disease detection to enhance agricultural diagnostics
- Modelled a sanctuary, for presentation, in **Free CAD** and **3D-printed** it in **PLA**, optimizing the design for **FDM** printing with proper wall thickness, layer height, and minimal supports

Machine Learning Developer

Traverse / Kathmandu, Nepal

December 2022 - May 2023

- Engineered a custom **recommendation engine** using **NumPy** and **PyTorch** to suggest places based on user itineraries.

Project Lead

NASA Space Apps Challenge / Kathmandu, Nepal

October 2022 - November 2022

- Developed an interactive 3D model of the International Space Station (ISS) and its orbit using **Three.js** and **JavaScript**

Technical Skills

OnShape, 3D Printing, Free CAD, Fusion 360, ESP32, RaspberryPi, Python, OpenCV-Python, PyTorch, FastAPI, JavaScript, ThreeJS, SvelteKit, Rust, C++, Machine Learning, Data Engineering, Git(hub), Database, Linux